



8-Bit Counter

Course: Digital IC Design and Tape out (ED221 & ED222)

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1. Introduction

1.1 Overview of CMOS Logic Gates and Sequential Elements

This experiment characterises five combinational gates — NOT, NAND, AND, OR, EXOR — and one sequential element, the D Flip-Flop (DFF). All modules are implemented at transistor level using the 180nm process at $V_{DD} = 1.8V$ in Cadence Virtuoso.

The DFF is implemented as a master-slave transmission-gate latch pair, providing edge-triggered behaviour. It serves as the storage element in counters, registers, and shift registers.

All modules in this experiment are implemented at the full custom transistor level using the TSMC 180nm CMOS process technology, with a nominal supply voltage of 1.8V. Transistor dimensions follow the minimum channel length of 180nm.

1.2 Why Timing, Power, and Energy Analysis Matters

In VLSI design, three parameters govern practical circuit performance:

- **Speed** — Gate delay limits the maximum clock frequency of any system built from those gates.
- **Power** — Excessive power increases heat, reduces battery life, and raises cost.
- **Energy efficiency** — A gate may be fast but wasteful, or efficient but slow. Combined metrics like PDP and EDP reveal the true cost of a design decision.

1.3 Key Definitions

Parameter	Symbol	Definition	Measured At
Propagation delay (HL)	t _{pHL}	Input 50% → Output 50% (falling)	50% of VDD
Propagation delay (LH)	t _{pLH}	Input 50% → Output 50% (rising)	50% of VDD
Average propagation delay	t _p	(t _{pHL} + t _{pLH}) / 2	—
Rise time	t _r	Output 10% → 90%	10%-90% VDD
Fall time	t _f	Output 90% → 10%	90%-10% VDD
Setup time (DFF)	t _{su}	Min. time D must be stable before clock edge	—
Hold time (DFF)	t _h	Min. time D must be stable after clock edge	—
Static power	P _{static}	$P = I_{\text{leakage}} \times V_{\text{DD}}$	No switching
Dynamic power	P _{dynamic}	$P = \alpha \cdot C_L \cdot V_{\text{DD}}^2 \cdot f$	During switching
Power-Delay Product	PDP	$P_{\text{avg}} \times t_p$	Energy per operation
Energy-Delay Product	EDP	$P_{\text{avg}} \times t_p^2$	Speed-energy trade-off

2. Modules Under Test

2.1 NAND Gate

Pin Description

Pin	Direction	Function
A	Input	Logic input 1
B	Input	Logic input 2
Y	Output	NAND output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Notes: No control pins. Both inputs must be driven to valid logic levels at all times. Floating inputs cause undefined output behaviour.

Schematic:

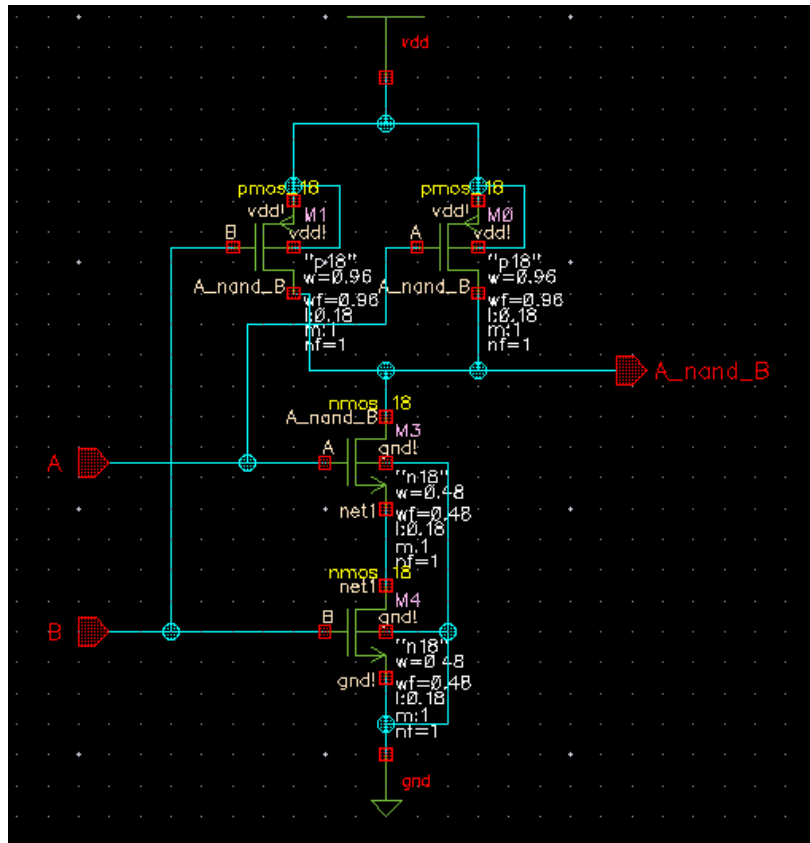


Figure 1: NAND Schematic

Symbol

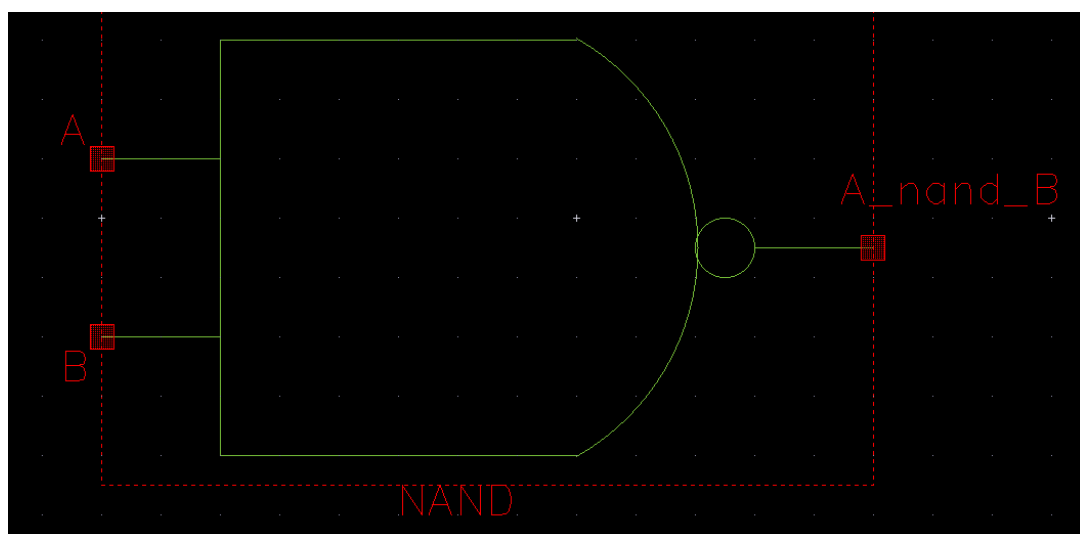


Figure 2: NAND Symbol

Truth Table

Boolean Expression: $Y = (A \cdot B)'$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

2.2 AND Gate

Pin Description

Pin	Direction	Function
A	Input	Logic input 1
B	Input	Logic input 2
Y	Output	AND output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Notes: No control pins. Implemented as NAND followed by NOT (inverter). Both inputs must be driven to valid logic levels.

Schematic

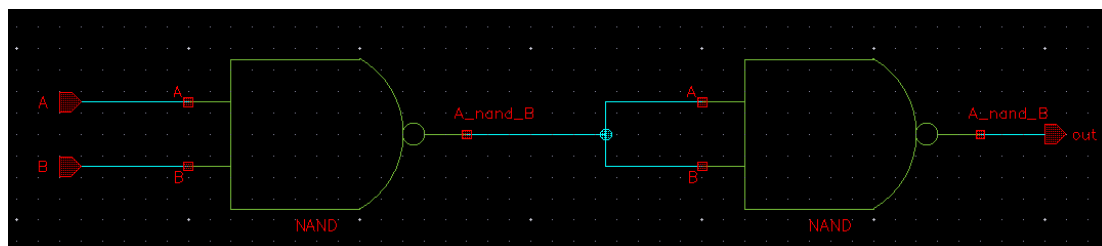


Figure 3: AND Schematic

Symbol

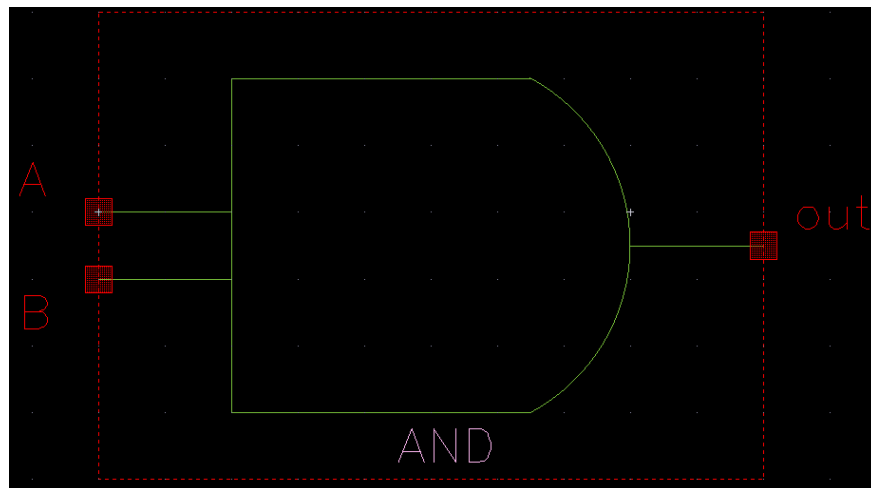


Figure 4: AND Symbol

Truth Table

Boolean Expression: $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

2.3 OR Gate

Pin Description

Pin	Direction	Function
A	Input	Logic input 1
B	Input	Logic input 2
Y	Output	OR output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Notes: No control pins. Implemented using NAND Gates.

Schematic

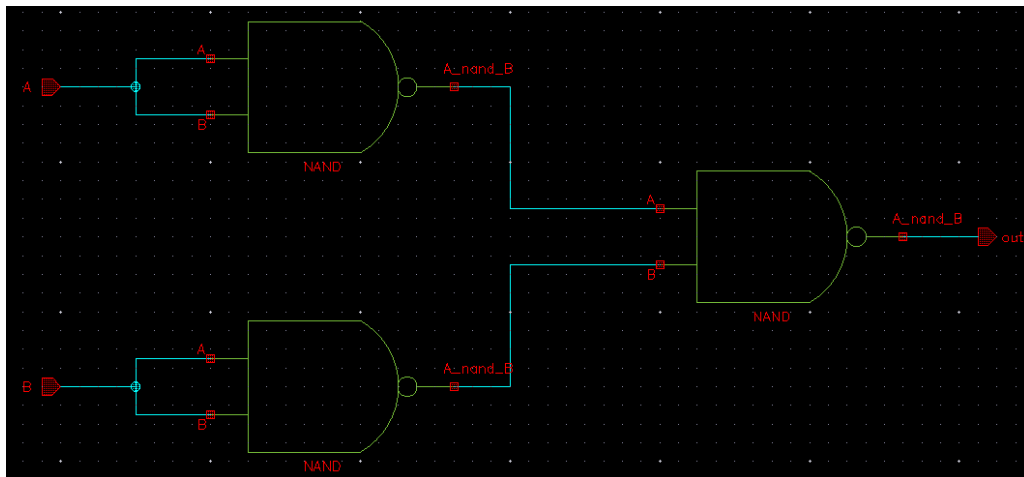


Figure 5: OR Schematic

Symbol

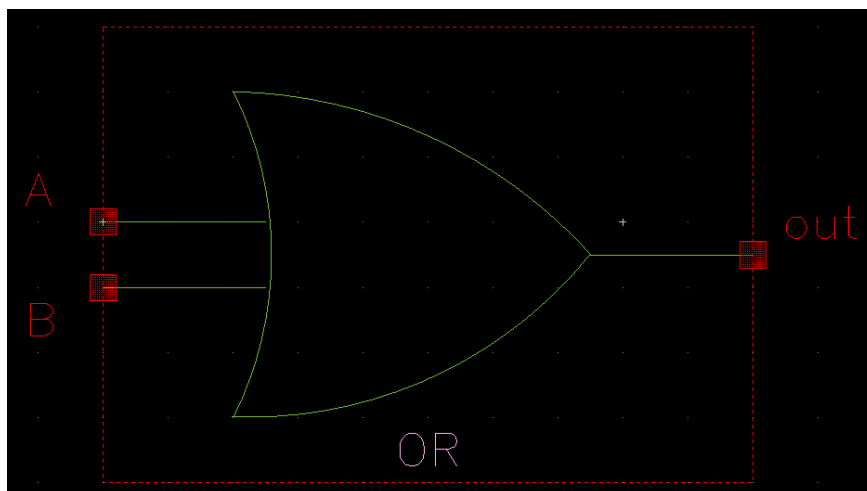


Figure 6: OR Symbol

Truth Table

Boolean Expression: $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

2.4 NOT Gate (Inverter)

Pin Description

Pin	Direction	Function
A	Input	Logic input
Y	Output	Inverted output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Notes: No control pins. Implemented using NAND Gate.

Schematic

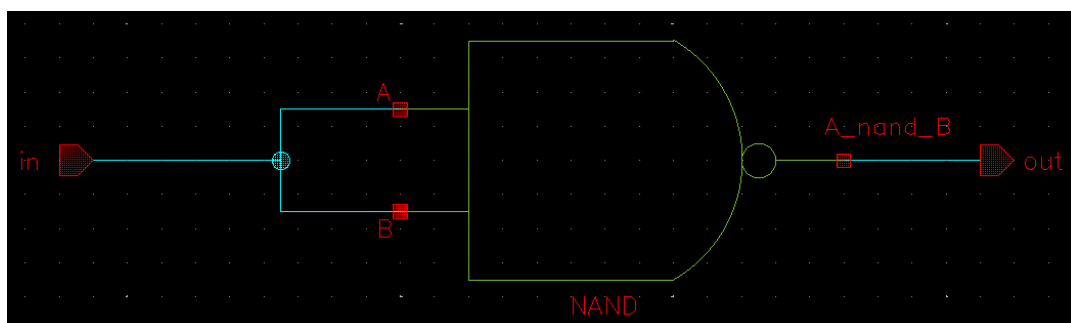


Figure 7: NOT Schematic

Symbol

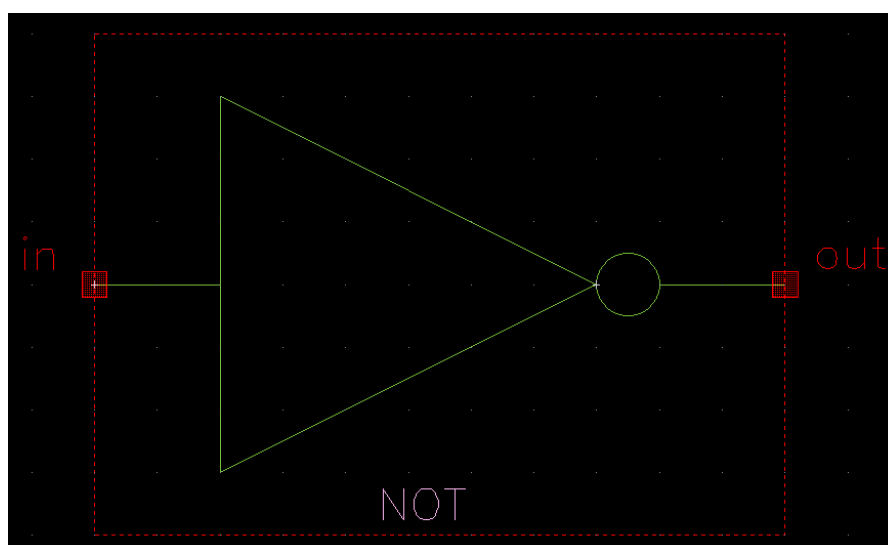


Figure 8: NOT Symbol

Truth Table

Boolean Expression: $Y = A'$

A	Y
0	1
1	0

2.5 EXOR Gate

Pin Description

Pin	Direction	Function
A	Input	Logic input 1
B	Input	Logic input 2
Y	Output	EXOR output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Notes: No control pins. Most complex gate in this experiment. Implemented using a combination of NAND/NOT gates or transmission-gate topology. Higher transistor count than other gates leads to higher capacitance and power.

Schematic

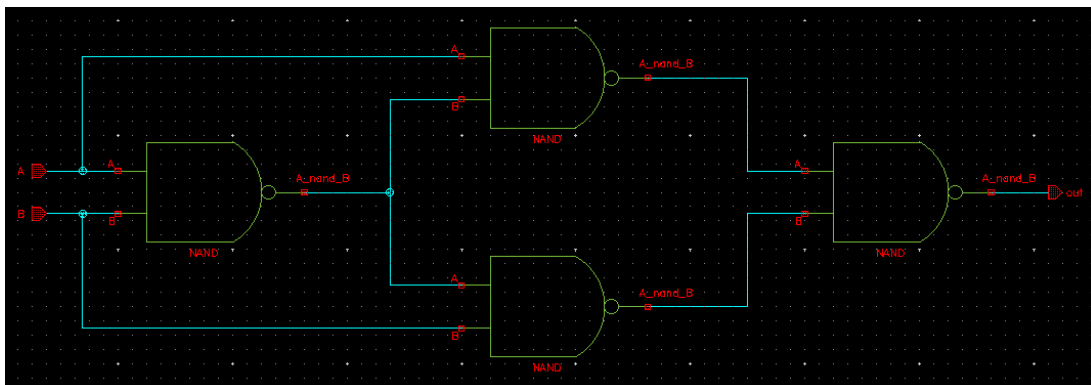


Figure 9: EXOR Schematic

Symbol

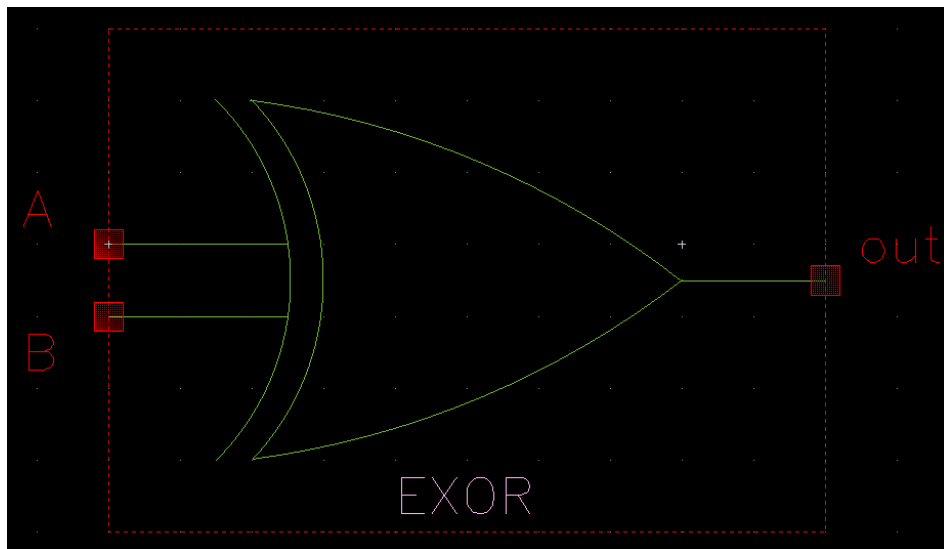


Figure 10: EXOR Symbol

Truth Table

Boolean Expression: $Y = A \oplus B = A'B + AB'$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

2.6 D Flip-Flop (DFF)

Pin Description

Pin	Direction	Function
D	Input	Data input
CLK	Input	Clock signal (rising edge triggered)
RST	Input	Synchronous active-high reset
Q	Output	Stored output
Q'	Output	Complementary output
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground reference (0V)

Ctrl Pin Usage & Notes

- **CLK:** The DFF captures the value of D on the **rising edge** of CLK only. The output Q holds its value between clock edges regardless of changes on D. All other modules in this experiment are combinational and have no clock pin.
- **RST:** When RST = 1, Q is forced to 0 synchronously on the next rising clock edge, overriding D. RST must be low (= 0) for normal operation.
- **Q and Q'** are always complementary. Driving both as outputs is useful for constructing T flip-flops and shift register stages.

Schematic

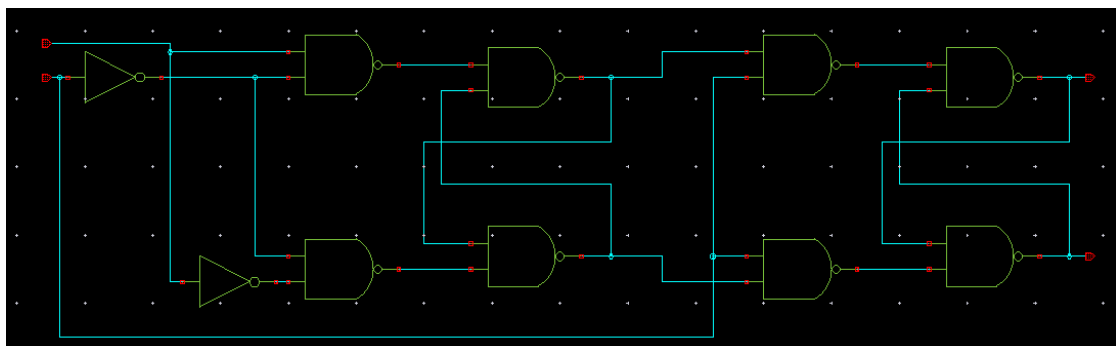


Figure 11: DFF Schematic

Symbol

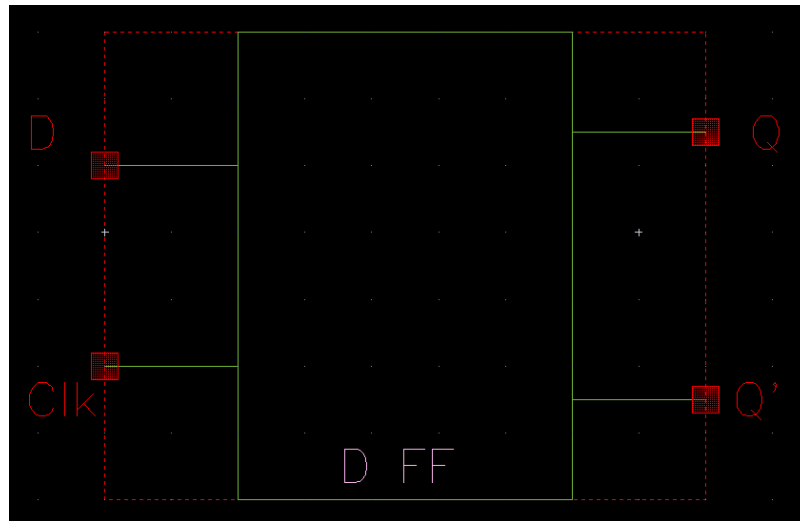


Figure 12: DFF Symbol

Characteristic Table:

RST	D	CLK	Q(next)
1	X	↑	0 (reset)
0	0	↑	0
0	1	↑	1
0	X	0 or 1	Q (hold)

2.7 8-Bit Synchronous Counter

Pin Description

Pin	Direction	Function
CLK	Input	Clock signal — counter increments on rising edge
RST	Input	Synchronous active-high reset — forces Q[7:0] = 0 on next rising edge
EN	Input	Count enable — counter increments only when EN = 1
Q[7:0]	Output	8-bit counter output, Q[7] = MSB, Q[0] = LSB
VDD	Power	Supply voltage (1.8V)
GND	Power	Ground (0V)

Ctrl Pin Usage & Notes

- **EN:** When EN = 0, the counter holds its current value regardless of CLK. When EN = 1, the counter increments by 1 on every rising CLK edge. Used to freeze the counter during the PISO shift phase.
- **RST:** Overrides EN. When RST = 1, Q[7:0] is forced to 00000000 synchronously on the next rising CLK edge.

Schematic

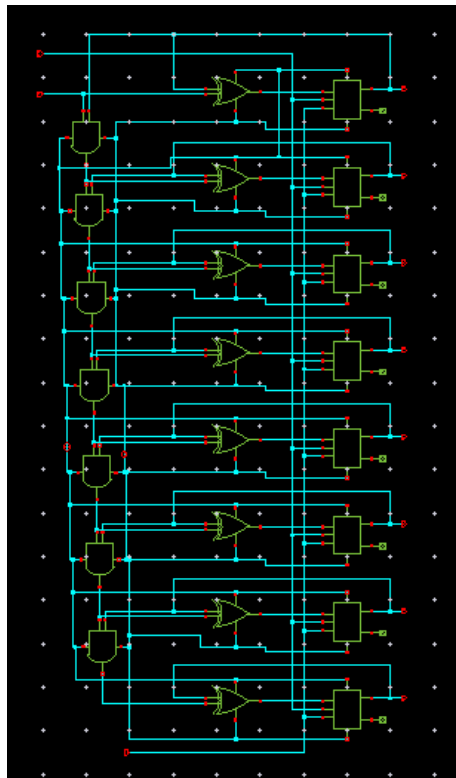


Figure 13: 8Bit Synchronous Counter Schematic

Symbol

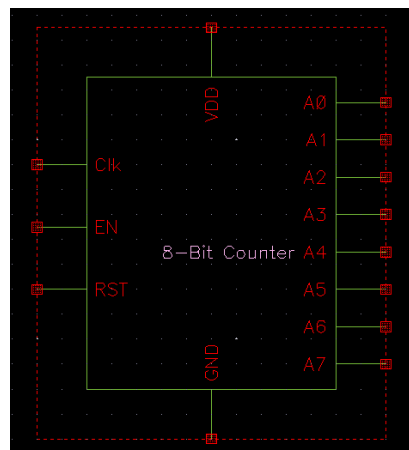


Figure 14: 8Bit Synchronous Counter Symbol

Working Explanation

The 8-bit synchronous counter is built from 8 D flip-flops connected with a ripple carry chain. Unlike an asynchronous (ripple) counter, all flip-flops share the same clock, ensuring simultaneous state updates with no ripple glitch propagation.

The carry logic determines when each stage toggles:

Stage	Toggle Condition
Q[0]	Every rising CLK edge (when EN = 1)
Q[1]	When Q[0] = 1
Q[2]	When Q[1] AND Q[0] = 1
Q[n]	When Q[n-1] AND ... AND Q[0] = 1

The counter counts from 00000000 (0) to 11111111 (255) and automatically rolls over to 00000000 on the next clock edge.

Output Waveform

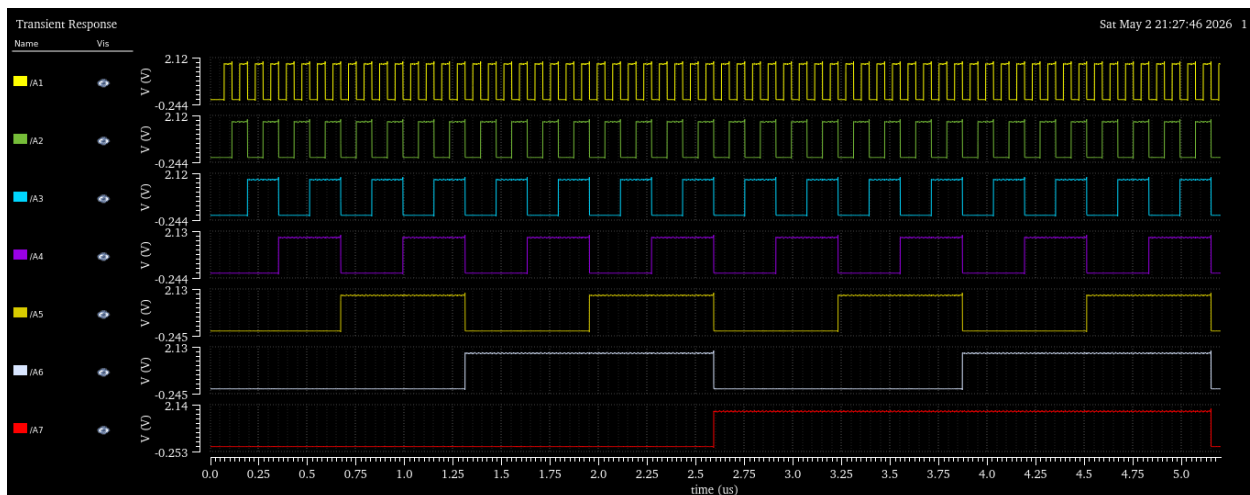


Figure 15: 8Bit Synchronous Counter output waveform

2.8 PISO Shift Register

Pin Description

Pin	Direction	Function
CLK	Input	Clock signal — shift occurs on rising edge
RST	Input	Synchronous active-high reset — clears all stages to 0
Shift_Load	Input	Mode: select — 0 = parallel load, 1 = serial shift
D[7:0]	Input	Parallel data input from counter Q[7:0]
DOUT	Output	Serial data output — MSB first
VDD	Power	Supply voltage (1.8 V)
GND	Power	Ground (0 V)

Ctrl Pin Usage & Notes

- **Shift_Load = 0 (Load mode):** On the rising CLK edge, all 8 stages simultaneously capture their respective D[i] inputs from the counter output. This is a single-cycle operation.
- **Shift_Load = 1 (Shift mode):** On each rising CLK edge, data shifts one position toward the output. The MSB (D[7]) exits first on DOUT, followed by D[6] down to D[0].
- **RST:** Clears all 8 internal stages to 0 synchronously, overriding both Shift_Load and D inputs.

Schematic

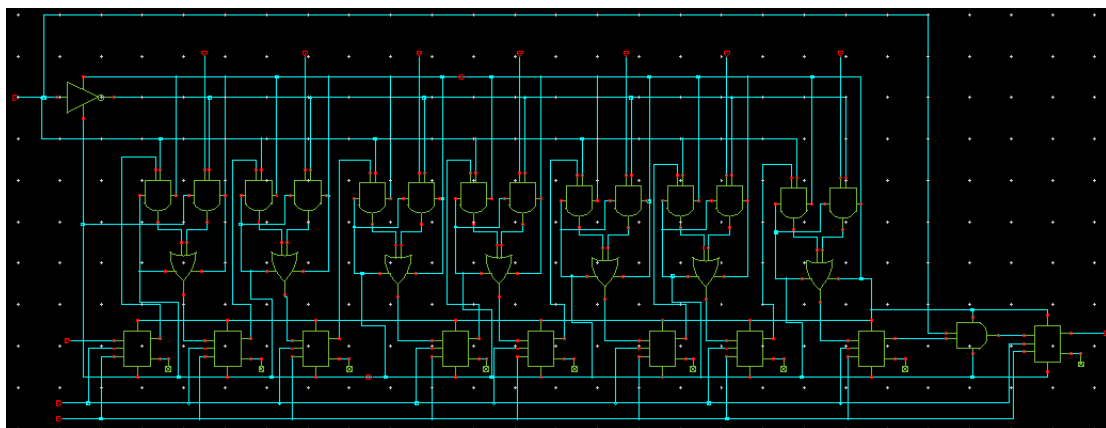


Figure 16: PISO Shift Register Schematic

Symbol

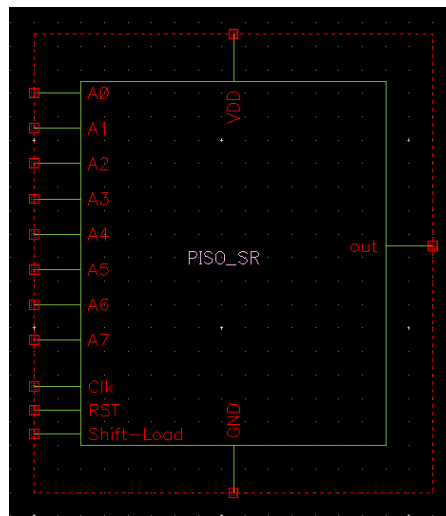


Figure 17: PISO Shift Register Symbol

Working Explanation

Shift_Load	MUX selects	Operation
0	D[i] from counter	Parallel load — all stages capture simultaneously
1	Q[i+1] from previous stage	Shift — data moves toward DOUT

Output Waveform

Sample input: 10110101

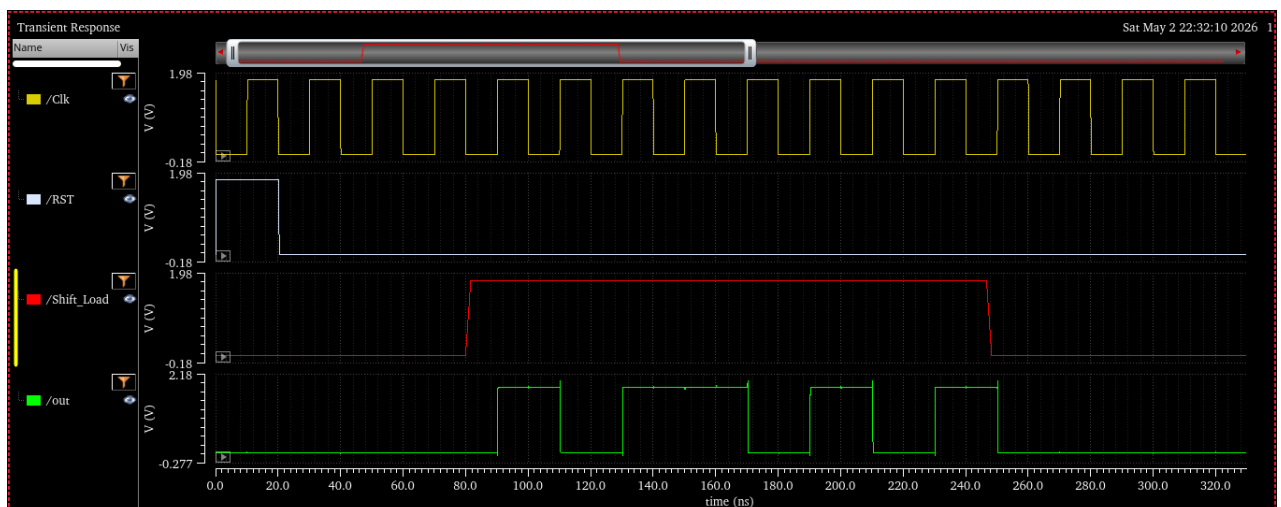


Figure 18: PISO Shift Register output waveform

2.9 Complete System Architecture

Top-Level Symbol

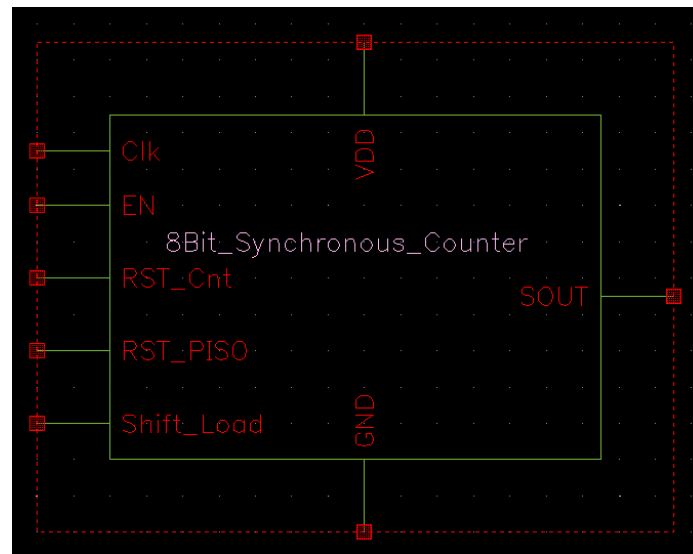
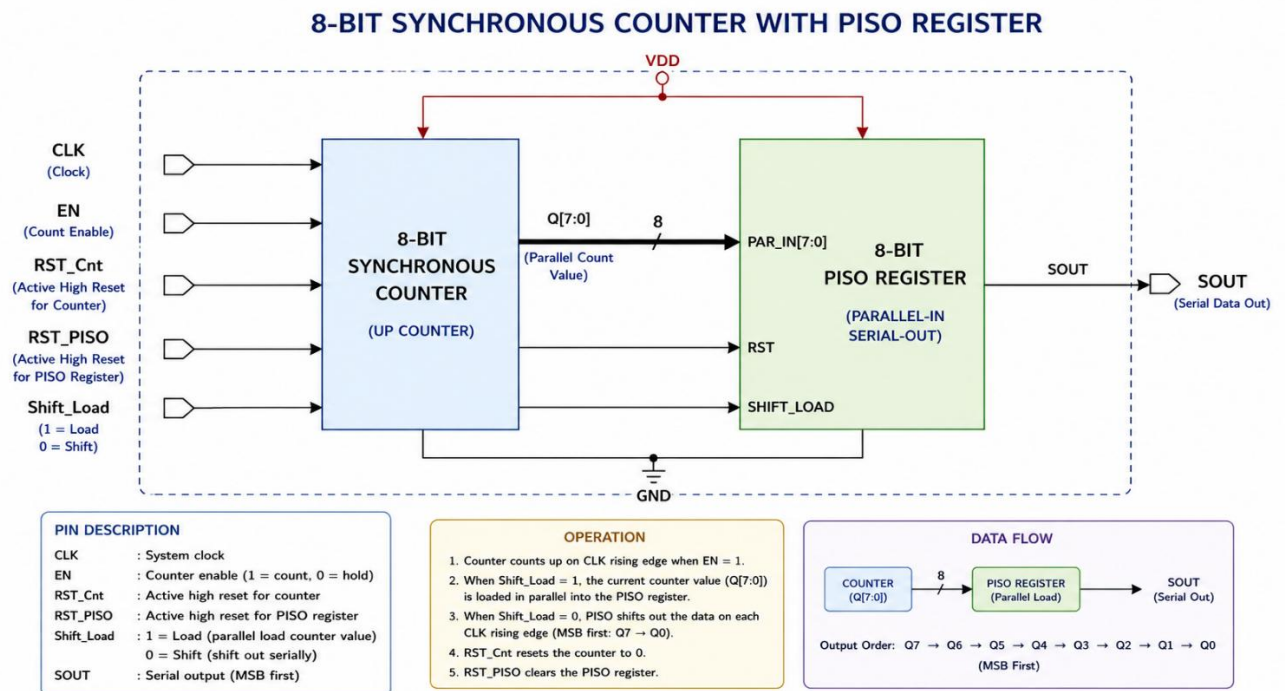


Figure 19: Complete System symbol

System Block Diagram



Pin Description — Complete Integrated System

Pin	Direction	Function
CLK	Input	Master clock — shared by counter and PISO
RST_C	Input	Synchronous reset for counter
RST_P	Input	Synchronous reset for PISO
EN	Input	Counter enable — high for 1 cycle to increment
Shift_Load	Input	PISO mode — 0 = load, 1 = shift
DOUT	Output	Serial output — MSB first, 1 bit per CLK
VDD	Power	Supply voltage (1.8 V)
GND	Power	Ground (0 V)

Signal Flow

The integrated system operates in a repeating 10-cycle frame:

1. **Count phase (1 cycle):** EN = 1 — counter increments by 1. Shift_Load = 1, PISO idle.
2. **Load phase (1 cycle):** EN = 0, Shift_Load = 0 — PISO captures counter Q[7:0] in parallel.
3. **Shift phase (8 cycles):** EN = 0, Shift_Load = 1 — PISO shifts out 8 bits on DOUT, MSB first. Counter held frozen.

The counter and PISO share CLK but have separate reset pins (RST_C, RST_P), allowing independent initialisation. Q0_MON provides direct visibility of the counter's LSB for debug and frequency verification without requiring a full serial readout.

3. Timing Analysis

3.1 NAND Gate

Timing Waveforms

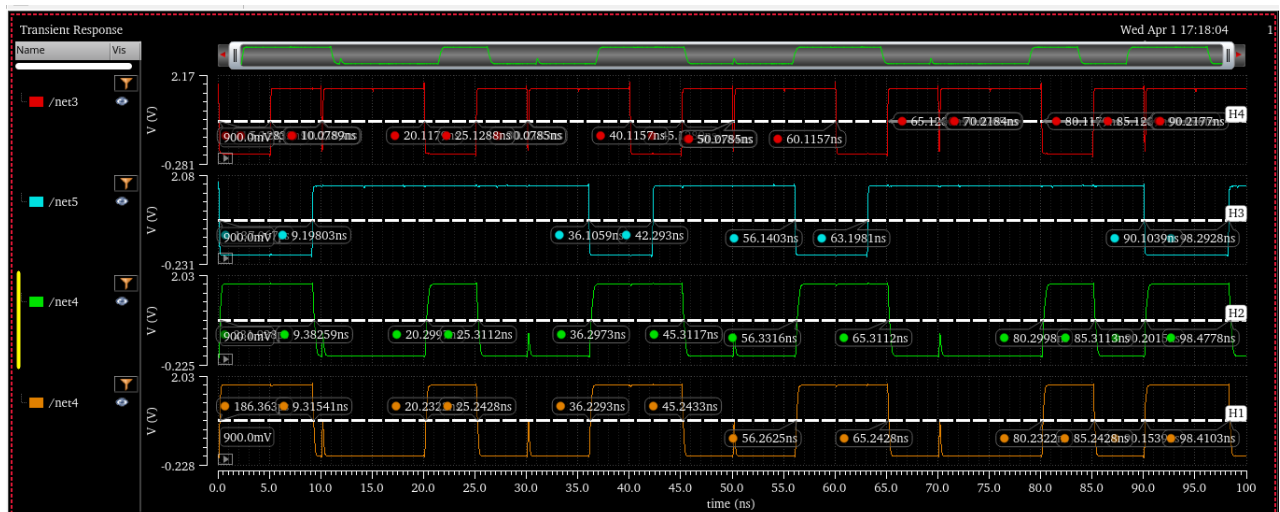


Figure 20: NAND Timing Waveform (Post Layout)

Propagation Delay

Measured from 50% point of input transition to 50% point of output transition.

Transition	Measurement	Value
T_{PHL}	Input LOW→HIGH, Output HIGH→LOW	0.10755 ns
T_{PLH}	Input HIGH→LOW, Output LOW→HIGH	0.13335 ns
T_P (avg)	$(T_{PHL} + T_{PLH}) / 2$	0.12045 ns

3.2 AND Gate

Timing Waveforms

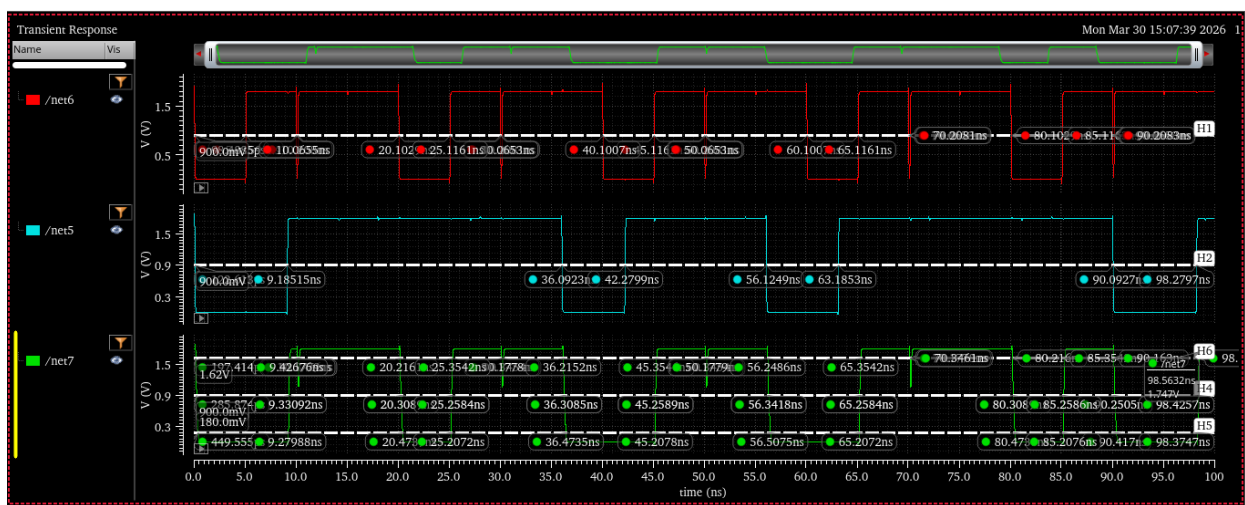


Figure 21: AND Timing Waveform

Propagation Delay

Transition	Measurement	Value
T_{PHL}	Input LOW→HIGH, Output HIGH→LOW	0.2111 ns
T_{PLH}	Input HIGH→LOW, Output LOW→HIGH	0.144155 ns
T_P (avg)	$(T_{PHL} + T_{PLH}) / 2$	0.1776275 ns

3.3 OR Gate

Timing Waveforms

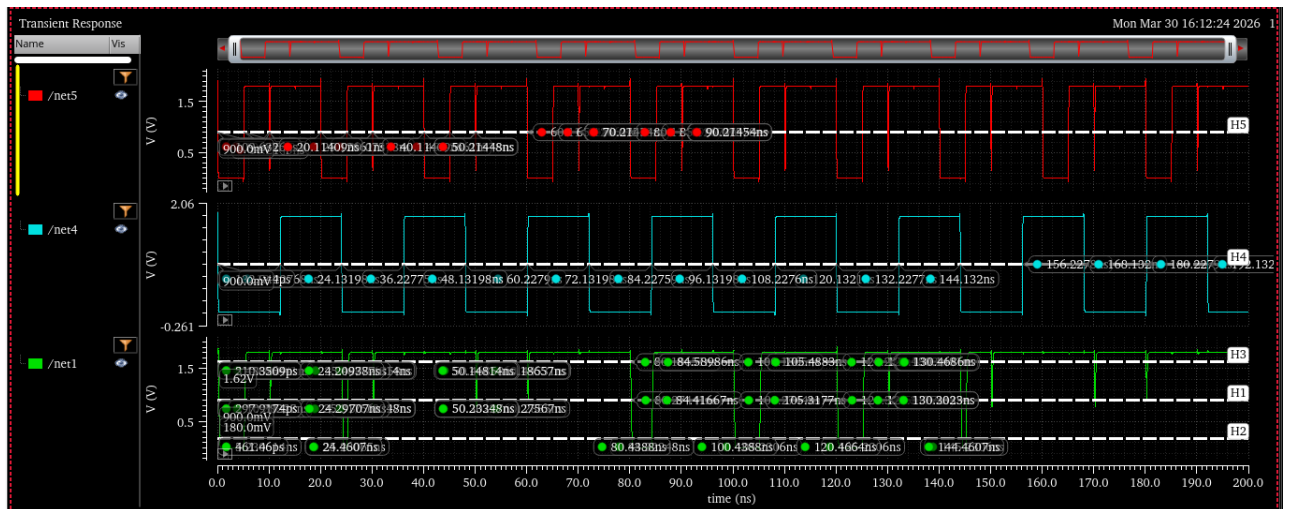


Figure 22: OR Timing Waveform

Propagation Delay

Transition	Measurement	Value
T_{PHL}	Input LOW $\rightarrow$ HIGH, Output HIGH $\rightarrow$ LOW	0.162695 ns
T_{PLH}	Input HIGH $\rightarrow$ LOW, Output LOW $\rightarrow$ HIGH	0.190408 ns
T_P (avg)	$(T_{PHL} + T_{PLH}) / 2$	0.17655 ns

3.4 NOT Gate (Inverter)

Timing Waveforms



Figure 23: NOT Timing Waveforms

Propagation Delay

Transition	Measurement	Value
T_{PHL}	Input LOW→HIGH, Output HIGH→LOW	0.0459 ns
T_{PLH}	Input HIGH→LOW, Output LOW→HIGH	0.0617 ns
T_P (avg)	$(T_{PHL} + T_{PLH}) / 2$	0.0538 ns

3.5 EXOR Gate

Timing Waveforms

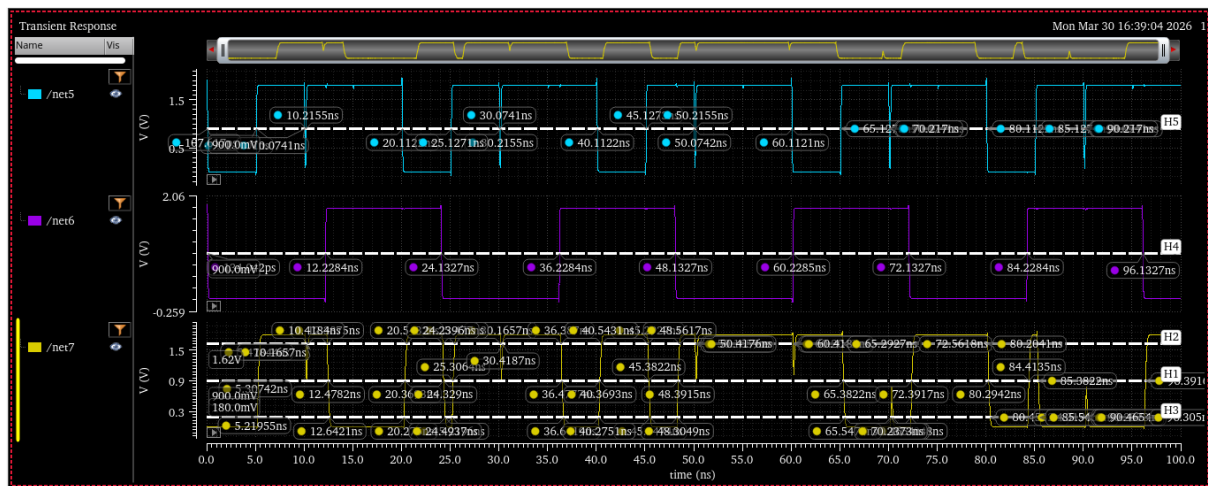


Figure 24: EXOR Timing Waveform

Propagation Delay

Transition	Measurement	Value
T_{PHL}	Input LOW→HIGH, Output HIGH→LOW	0.22034 ns
T_{PLH}	Input HIGH→LOW, Output LOW→HIGH	0.22891 ns
T_P (avg)	$(T_{PHL} + T_{PLH}) / 2$	0.22462 ns

3.6 D Flip-Flop (DFF)

Timing Waveforms

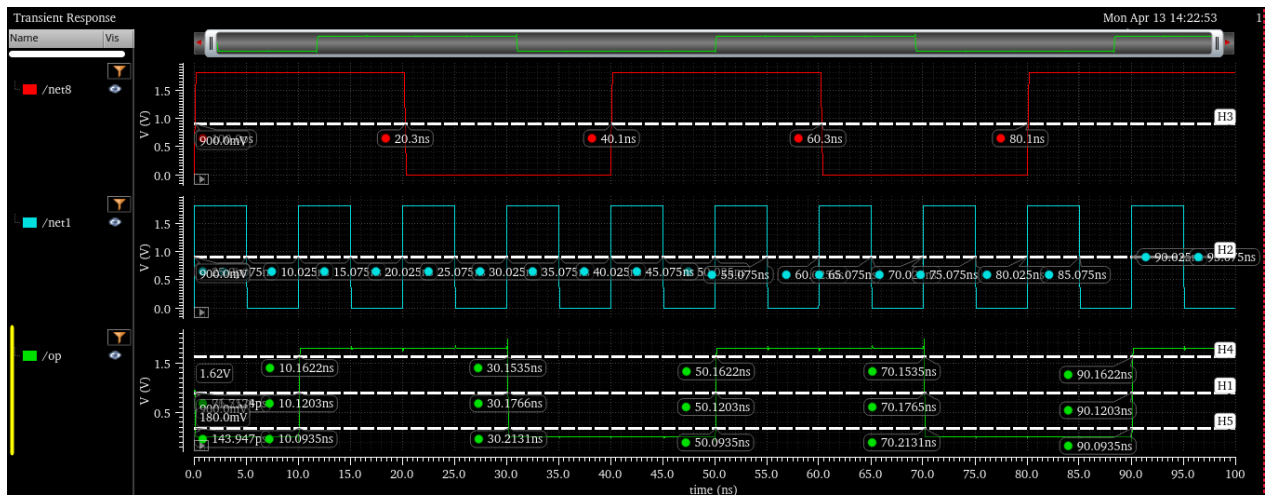


Figure 25: DFF Timing Waveform

Propagation Delay

Transition	Measurement	Value
T_{PHL}	Input LOW→HIGH, Output HIGH→LOW	0.15155 ns
T_{PLH}	Input HIGH→LOW, Output LOW→HIGH	0.0953 ns
T_{C-Q}	$(T_{PHL} + T_{PLH}) / 2$	0.1328 ns

4. Power & Energy Analysis

4.1 Measurement Methodology

Power was calculated from the average current drawn from the VDD supply during transient simulation in Cadence ADE-L, with the clock always running:

$$P = I_{avg} \times VDD$$

4.2 Current and Power Measurement

Parameter	Condition	Value
Average I	CLK running at 50 MHz, full system active	40.36uA
Max I	CLK running at 50 MHz, full system active	5.7889 mA
Average Power	$P = I_{avg} \times 1.8 \text{ V}$	72.64uW
Max Power	$P = I_{max} \times 1.8 \text{ V}$	10.4199mW

4.3 Energy per Operation

Energy per clock cycle:

$$E_{\text{cycle}} = P \times T_{\text{clk}} = P \times 20 \text{ ns}$$

Parameter	Value
E_{cycle}	1.4528 pJ

Energy per complete frame (one full 8-bit value shifted out = 10 clock cycles):

$$E_{\text{frame}} = P \times 10 \times T_{\text{clk}} = P \times 200 \text{ ns}$$

Parameter	Value
E_{frame}	14.528 pJ

6. Complete End-Result Timings

6.1 System Timing Waveforms

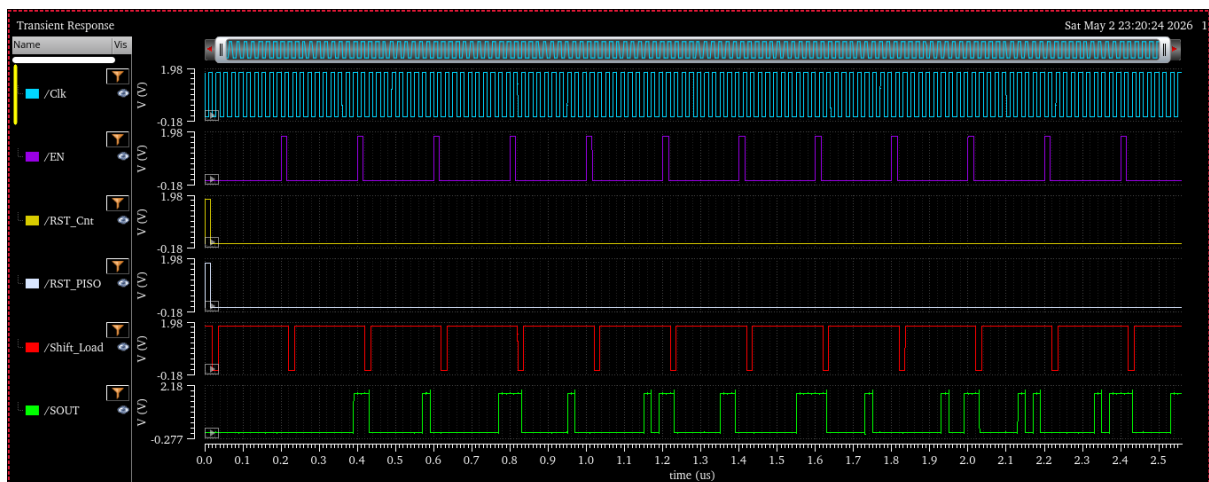


Figure 26: 8Bit Counter + PISO timing waveforms

6.2 Frame Timing

One complete frame = 10 clock cycles as follows:

Phase	Signal Condition	Duration	Purpose
Count	EN = 1	1 cycle = 20 ns	Counter increments by 1

Load	EN = 0, Shift_Load = 0	1 cycle = 20 ns	Counter value captured into PISO
Shift	EN = 0, Shift_Load = 1	8 cycles = 160 ns	8 bits shifted out on DOUT
Total	—	10 cycles = 200 ns	One complete value output

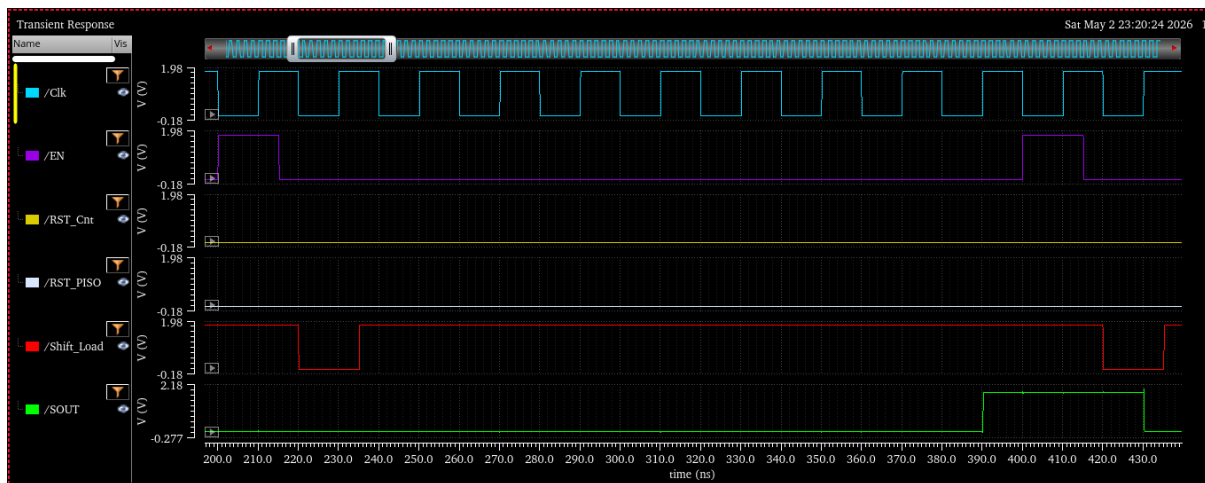


Figure 27: One complete frame

6.3 System Timing Measurements

CLK to DOUT delay — time from CLK rising edge to DOUT settling:

Parameter	Value
CLK → DOUT (first bit after LOAD)	0.13 ns

6.4 Observations

- The counter increments by exactly 1 between consecutive frames, confirming correct synchronous counting behaviour.
- DOUT correctly outputs the LSB first over 8 clock cycles, matching the parallel-loaded counter value.
- No glitches are observed on DOUT between bit transitions, confirming the registered output DFF is functioning correctly.

- The system operates without timing violations at 50MHz, confirming adequate timing margin over the critical path.
- EN and LOAD separated by one clock cycle eliminates the race condition, producing consecutive output values 0, 1, 2, 3...

7. Conclusion

This experiment involved the transistor-level design, implementation, and simulation of fundamental CMOS logic gates (NOT, NAND, AND, OR, EXOR), a D flip-flop, an 8-bit synchronous counter, and a PISO shift register using the 180nm process at $V_{DD} = 1.8V$ in Cadence Virtuoso.

7.1 Summary of Findings

Timing:

- The NOT gate exhibited the lowest propagation delay among all modules, confirming that simpler gate topology directly translates to faster switching.
- The NAND gate was the fastest two-input gate, consistent with the CMOS property that parallel PMOS pull-up networks are more efficient than series configurations.
- AND and OR gates showed higher delays than NAND and NOR respectively, due to the additional inverter stage in each implementation.
- The EXOR gate had the highest propagation delay among combinational gates, reflecting its larger transistor count and internal capacitance.
- The DFF showed the highest overall delay due to its master-slave two-latch structure and the additional setup time constraint on the data input.
- t_{pHL} was consistently shorter than t_{pLH} across all modules, confirming the higher drive strength of NMOS transistors relative to PMOS in the 180nm process.

Power and Energy:

Total system power of the integrated 8-bit counter + PISO was measured as 72.64 μW at 50MHz, computed as $P = I_{avg} \times V_{DD}$

- Energy per clock cycle was 1.4528 pJ and energy per complete 8-bit output frame (10 cycles) was 14.528 pJ.
- Dynamic power dominated due to continuous clock activity across all 16 flip-flops, the carry chain, and PISO MUX logic.

System Performance:

- The integrated system operated correctly at 50MHz ($T_{\text{clk}} = 20\text{ns}$), producing consecutive 8-bit values 0, 1, 2, 3... on DOUT in MSB-first serial format.
- The 10-cycle frame architecture (1 count + 1 load + 8 shift) successfully resolved the pin constraint, reducing 8 parallel output lines to a single DOUT pin within the 6-pin functional budget.
- Separating EN and Shift_Load by one clock cycle eliminated the race condition that previously caused incorrect output values.